

Universal Exponent $\beta = 3/2$ in CSGT

Derived from Dimensional Holographic Scaling of Coherence Field Fluctuations

$$\beta = 1 + \frac{1}{D-2} \quad \Rightarrow \quad D = 4 \quad \Rightarrow \quad \beta = \frac{3}{2}$$

Derivation basis: Holographic equipartition near horizon
 $\langle \delta C \delta C \rangle \propto T_H^{2/(D-2)}$
(C-field quantum fluctuation scales with surface gravity $\kappa \sim T_H$)



CMB Phase Shift

Information lag from $C(z)$ evolution

$$\Delta\phi \propto \int C(z) dz \sim \beta \cdot \Delta z$$

Observed: $\Delta\ell \approx +1.48$ at $z \approx 1100$

(late-universe coherence recovery)

C relaxation

Black Hole Evaporation & Page Curve

Near-horizon: $\langle \delta C \delta C \rangle \sim T_H^{2/(D-2)}$

Late-time info release rate $\propto t^{-\beta}$

Page curve: backflow driven by future $C \rightarrow 1$
(no replica/island needed)

C relaxation



GW Ringdown Non-Exponential Decay

$$h(t) \sim \exp[-(t/\tau)^\beta] \quad \text{with } \beta = 3/2$$

Relaxation time $\tau \sim 1/\kappa \sim 4GM$

(independent of M, J, Q in late tail)

κ : surface gravity $\sim T_H$

All phenomena unified by C -field quantum relaxation

$\rightarrow \beta = 3/2$ as the quantum gravity fingerprint in 4D spacetime

Caption: The exponent $\beta = 3/2$ arises inevitably from holographic equipartition and quantum fluctuations of the coherence field C in 4-dimensional spacetime.

It governs information recovery across cosmology (CMB phase shift), black hole physics (Page curve & late evaporation), and gravitational waves (ringdown tail), offering a model-independent, falsifiable signature of CSGT as a unified gravity-quantum framework.